

SAP Master Data Governance 9.0

Central governance of material data

June 2016



Objectives Slide



At the end of this lesson, you will be able to:

- Explain the goal of Master Data Governance for Material
- Describe the strategy and the features of the current solution
- Outline the architecture of Master Data Governance for Material
- Describe the governed maintenance process for material master data with Change Request and workflow



Agenda

1. **Current Reality**
2. Need for the Future
3. SAP Master Data Governance Capabilities
4. Architecture
5. Current Solution
6. Benefit from the Solution



Questions From the Manufacturing and Logistics LOB

Material Governance



- How can I establish a “maintain once – use everywhere” best practice?
- How can I reduce cost by eliminating redundant effort to maintain master data, i.e. through avoiding duplicates?
- How can I increase the quality of my material master data (i.e. classifications etc.) through central management and governance?
- How can I get consistent insight into sales, manufacturing, and procurement volumes?
- How can I accelerate availability of up-to-date master data in my business network?
- How can I get transparency on who has changed what, when and why?
- And ... while accomplishing all that ... how can I best leverage my investment in SAP?

- How can I establish a “maintain once – use everywhere” best practice? → Maintenance in one central MDG(M) hub
- How can I reduce cost by eliminating redundant effort to maintain master data, i.e. through avoiding duplicates? → one source of truth, duplicate search,
- How can I increase the quality of my material master data (i.e. classifications etc.) through central management and governance? → integrated into business logic (customizing and data verification)
- How can I get consistent insight into sales, manufacturing, and procurement volumes? → use of consistent data not only in operational but also business intelligence context
- How can I accelerate availability of up-to-date master data in my business network? → workflow-driven processing and (semi-)automated distribution
- How can I get transparency on who has changed what, when and why? → central change request handling and process analytics on the hub

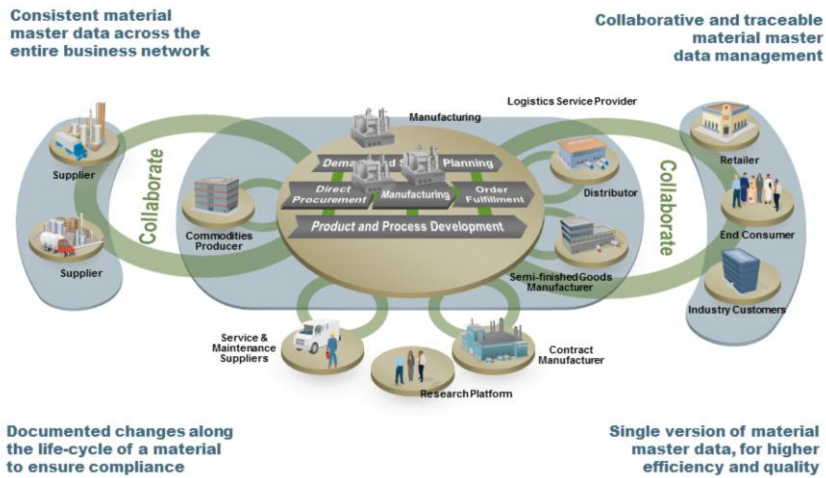


Agenda

1. Current Reality
- 2. Need for the Future**
3. SAP Master Data Governance Capabilities
4. Architecture
5. Current Solution
6. Benefit from the Solution



Central Management of Material Master Data Drives Excellence Throughout Your Business Network



Single Entry Point to maintain material data → consistent material master data across the entire business network

Standardized Process to control data changes → documented changes along the life-cycle

Standardized Tools for ensuring data quality → single version of the truth

Data Flow Transparency within system landscape → collaborative and traceable

Get a Snapshot of the relevant material related activities

Transparency on the data changes done (Who, When, What, Why)



Agenda

1. Current Reality
2. Need for the Future
- 3. SAP Master Data Governance Capabilities**
4. Architecture
5. Current Solution
6. Benefit from the Solution



What is SAP Master Data Governance for Material Data?

SAP Master Data Governance Capabilities

Master Data Governance for material data offers a central hub-based maintenance of material master data.

To address the key business issues related to such a governance solution for master data, MDG for material follows some basic principles that are shared across the domains of MDG

- Maintain once, use many
- One standardized process to be used by everybody
- One standardized user interface for everybody
- One standardized set of business logic

In order to fulfill these principles, SAP has built a Master Data Governance solution that offers

- Central based maintenance of master data
- Workflow-driven processes that are flexible while ensuring governance and standardization
- Harmonized user interface, based upon Enterprise Portal or the Business Client
- Data model and business logic follow the best practices of the SAP Business Suite and reuse existing data structures, business validations and various service

SAP Master Data Governance Capabilities

Establish Best Quality Master Data to Optimize Business Processes



I would now like to take you through some of the product's capabilities, that we believe make MDG a very attractive offering – especially for our Business Suite customers. In the course of the next couple of slides we will discuss questions like: What does “out-of-the-box” mean in this context? What are domain-specific applications that come with an “Integrated Object Model”? How do we enable Governance in a way that allows for collaboration and better data quality? How do we support data replication? And how do we allow customers to re-use our (and their own) past investments and do we allow them to extend the standard?

SAP Master Data Governance Capabilities

1 Deliver Out-of-the-Box

	Low TCO	Full Integration to SAP Systems as a client Offer integration services for non-SAP client systems
	Flexibility	Workflow flexibility closely integrated to a framework for defining business rules
	Ease of Consumption	Deliver a process-centric solution, with roles and a web-based user interface Deliver predefined data quality services
	Best Practice	Deliver a data model proven by several applications & industries
	Process Transparency	Reuse of existing processing logic
	Integration	Out-of-the box data replication mechanisms, using stable mechanisms

© 2016 SAP SE or an SAP affiliate company. All rights reserved.

Early Knowledge Transfer

Customer

10

What does “out-of-the-box” meant in the context of MDG?

Our customers regularly complain about the high cost for implementing and running software solutions. This is also an issue when they are investigating MDM Solutions of our competitors. With SAP MDG we have a clear advantage here, as we offer MDG in way that you can for example co-deploy it with existing SAP ERP system.

MDG is easy to consume and ready to run. It comes with standard roles, workflows, pre-defined workflow inboxes for processing change requests, with user interfaces for data maintenance or approval, interfaces for integration to BusinessObjects Data Services and so on. This allows customers to immediately start with a proof-of-concept. And it avoids one of the probably most costly parts of any other MDM implementation – the reuse of existing processing logic in the back-end.

But regardless of its out-of-the-box nature, at the same time MDG is very flexible and this flexibility comes at a low price tag. There is many things you can change by configuration: for example by adapting workflows, or validation rules in BRF+. For many of this adaptations you do not need to change or create any coding.

MDG’s pre-delivered “best practice” data models are based on the models in the SAP Business Suite and hence are already proven in the SAP business context across multiple applications and industries.

To a large extend, MDG behaves like master data applications in the Business Suite already do, especially in the sense that MDG uses the exact same validations that existing master data transactions use.

In order to replicate the data from MDG into other systems, it uses established replication mechanisms (like existing IDoc templates) and thus provides out-of-the-box integration mechanism into the Suite, but of course also the means to replicate to non-SAP systems (for example using Enterprise Services).

SAP Master Data Governance Capabilities

2 Integrated Object Model

Identifying Material Data ~ 10 fields

- Both SAP internal (material number) as well as external standards (GTIN)

Descriptive Material Data ~ 100 fields

- Help to understand what a material is but do not control processes
- Are as different as the products of our customers
- Typically realized via Classification (rather than Field Extension)

Process Controlling Material Data ~ 1.000 fields

- Control the behavior of SAP business processes
- Proprietary to SAP applications



MDG 9.0 Scope includes (~ 500 fields)

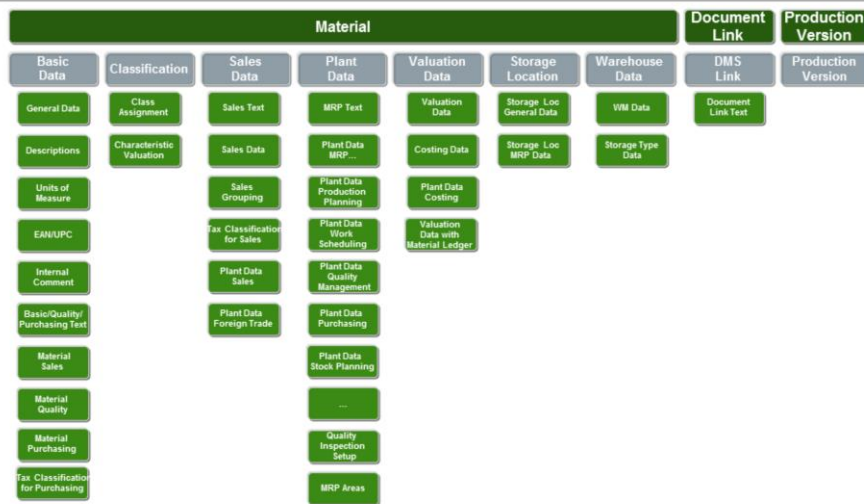
- Basic Data and Long texts / notes
- Quantity Conversion / alternative UOMs/GTIN / EAN handling
- Classification
- Plant specific data (with for Inspection Setup, MRP Area)
- Valuation data (with Material Ledger)
- Sales Organization / Distribution Channel specific data / Tax Classification
- Storage Location Data
- Warehouse Data
- Document Assignment
- Production Version

Customers can enhance the data model

SAP Master Data Governance Capabilities

2

Integrated Object Model



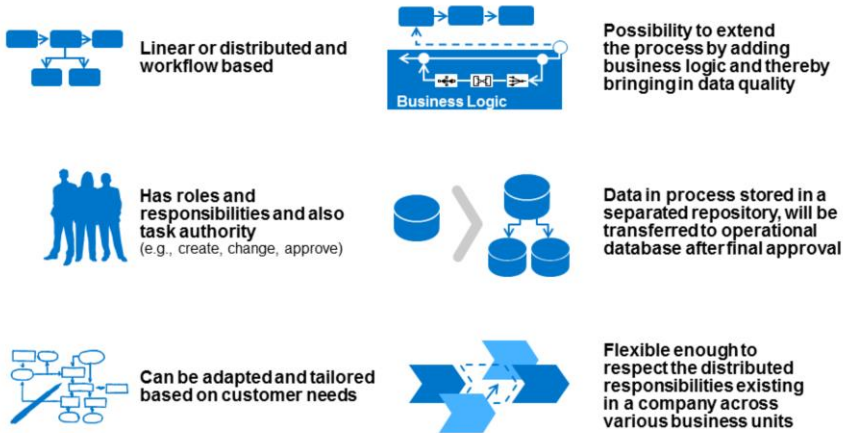
This slide is more intended to be a quick reference for you and an overview about the current standard data models that we deliver with MDG in the release SAP MDG 8.0. Later in this course we will cover the master data domains in MDG one by one.

Just a few words here: SAP MDG in its current version covers the domains Financials, Material, Supplier, and Customer. MDG for Financials Data is covering Accounting data like Chart of Accounts, General Ledger Accounts and their Hierarchies, Financial Reporting Structures, Controlling data like Cost Elements and their Hierarchies, Profit and Cost Centers and their Hierarchies, and Companies, as well as Financial Consolidation data like Items and Groups with their respective Hierarchies, Units and Break Down Categories. For Material we now also have a very broad model that covers close to 400 attributes. This includes General Data, text and description fields, as well as units of measure and tax information and classification. In addition, we have valuation and costing data as well as logistics attributes that are dependent on organizational units like sales organizations, distribution channels, or plants, as well as Storage and Warehouse data. The Supplier model is based on a broad range of generic business partner attributes incl. addresses and also business partner relationships. In addition to that it contains ERP specific supplier attributes like Purchasing Data or Company Code Data. Also the Customer model is based on the generic business partner and contains its attributes. Same as the supplier, in addition to that the customer model also contains ERP specific customer attributes like Sales Area Data, Company Code Data and Tax Indicators.

SAP Master Data Governance Capabilities

3

Governance, Collaboration...



Part of the solution strategy are principles that represent the people-centric and flexibility goals of the solution.

This is not specific to material data, but holds true for the overall SAP MDG solution.

In general the processes of the solution can be adapted and tailored to the customer needs.

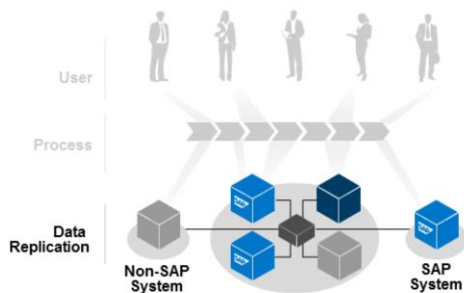
This includes:

- Workflows that are even linear, can be distributed or even started ad hoc
- Clear roles and responsibilities following a task-oriented approach
- Process steps can be added, either reusing existing logic or using a flexible rules-engine
- The data in the governance process is separated but transfer to operational database is done in very controlled mode

In summary this fosters Flexibility and User Acceptance while ensuring Governance

SAP Master Data Governance Capabilities

4 Robust Data Replication



Replication

- Direct, pooled or manual replication to SAP and non-SAP systems
- Peer-to-peer or mediated

Replication Techniques

- DRF controlling replication mechanisms like
 - ALE for SAP ECC, R/3 and non SAP systems
- CIF for SAP SCM and MW for SAP CRM/SRM

Replication Features

- Flexibility to replicate selected data only to a specific client system by defining filters (controlled by DRF)
- Support of different object IDs or customizing keys between hub and client systems (via mapping in SAP MDG or SAP PI)
- Monitoring and error handling

Once you have maintained and approved the master data on the central MDG system, you might want to replicate the data to all SAP and non-SAP systems that will need the master data. MDG comes with a framework that supports the organization and execution of this data replication – it is called Data Replication Framework (DRF).

With MDG, we of course did not want to invent yet another way to do this replication, but we intended to better support customers in managing the distribution they already do today. Accordingly the DRF framework sits on top of the established replication mechanisms that SAP customers have already been using for a long time (like ALE IDoc) and newer ones like Enterprise Services. The purpose of MDG's DRF is to help our customer keeping track of what needs to be distributed where, and what has been distributed successfully or not – regardless of the various replication mechanisms used underneath. In DRF, systems can subscribe to the distribution of master data based on certain filter criteria – like only materials for plant 1000 with material type FERT are distributed to system XYZ.

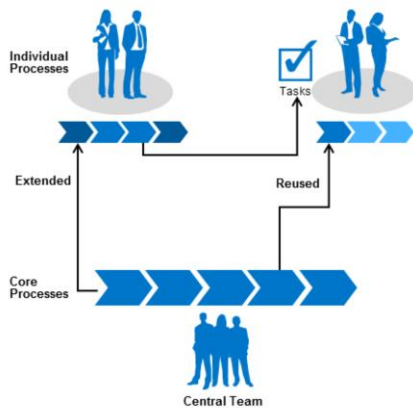
In addition MDG can handle things like key mapping (in order to keep different historical keys for existing master data in de-central systems and at the same time only maintain the data once centrally). It also manages value mapping (in order to allow distribution into systems that might have a different understanding of the values of an attribute, for example material type FERT in one system might be called PROD in other systems).

DRF also helps in monitoring the distribution, and in resolving errors.

SAP Master Data Governance Capabilities

5

Reusability & Extensibility



Reuse standard content

- Reuse existing data structures and functionality of SAP Business Suite

Enhance functionality

- Be able to enhance MDG functionality based on predefined content (e.g. plant or sales data) or customer-specific extensions

Extend data structures

- Provide easy-to-use extensibility capabilities for enhancing nodes and fields
 - data models
 - user interfaces
 - processes
 - data replication

A major focus of MDG has been put on re-use and extensibility.

But re-use does not only mean re-use for us in SAP development. Yes, we are re-using the existing ERP coding to validate entered data and we are often re-using the existing tables to store the active data after approval.

But more importantly it is also about re-use for our customers: They can re-use extensions that they have already made to SAP master data in the Business Suite and they can include these extensions directly in the MDG data models. They can re-use the own business logic that they have created in SAP Business Suite for example to validate newly created master data.

We know that each and every SAP customer has made extensions to SAP's standard master data model. And that many companies have very specific master data creation and change processes. Accordingly, MDG offers the flexibility and extensibility features to adapt the standard to their needs. The big value proposition here is that we allow our customers to do these extensions and still benefit from MDG out-of-the-box capabilities.



Agenda

1. Current Reality
2. Need for the Future
3. SAP Master Data Governance Capabilities
- 4. Architecture**
5. Current Solution
6. Benefit from the Solution



Principles



User-Centricity

- Central Work Center for the Maintenance of Materials
- The work center for material master data governance bundles together all activities for material master data maintenance.
- Central deployment and authorization ensures governance
- Local adaptation and personalization possible



Governance

- Change Requests With Built-In Approval Process
- Integrated process starts with a request to create or change a material, continues with processing and commenting by different expert users and stakeholders, and finishes with approval or rejection by authorized experts
- All processing steps are supported by workflow



Workflow

- Standardized but Flexible Processes Incl. Rules Management and Tracking
- The standard SAP Business Workflow is enhanced with a rule-based engine; therefore changes in responsibilities and in processes can be reflected immediately and with ease.



Architecture

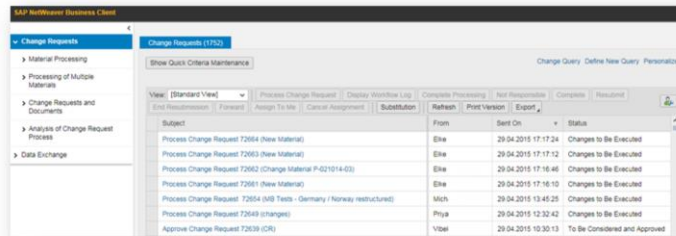
- Staging Area to maintain inactive data of Material
- Active area – Holds the active data; Staging – Holds the data during the change request processing before approval

User-Centric Principle

Central Work Center for the Maintenance of Materials

All activities around the creation and maintenance of materials are bundled in one work center. Companies can establish this work center as the standard hub for creating and changing materials.

For customers that do not use the SAP Enterprise Portal, a work center based on SAP Netweaver Business Client is delivered as well.



Subject	From	Sent On	Status
Process Change Request 72654 (New Material)	Elna	29.04.2015 17:17:24	Changes to Be Executed
Process Change Request 72653 (New Material)	Elna	29.04.2015 17:17:12	Changes to Be Executed
Process Change Request 72652 (Change Material P-021014-03)	Elna	29.04.2015 17:16:46	Changes to Be Executed
Process Change Request 72651 (New Material)	Elna	29.04.2015 17:16:10	Changes to Be Executed
Process Change Request 72654 (MS Tests - Germany / Norway restructured)	Mith	29.04.2015 13:45:25	Changes to Be Executed
Process Change Request 72649 (changes)	Priya	29.04.2015 12:32:42	Changes to Be Executed
Approve Change Request 72639 (CR)	Vibaj	29.04.2015 10:30:13	To Be Considered and Approved

- **Customer business challenges**

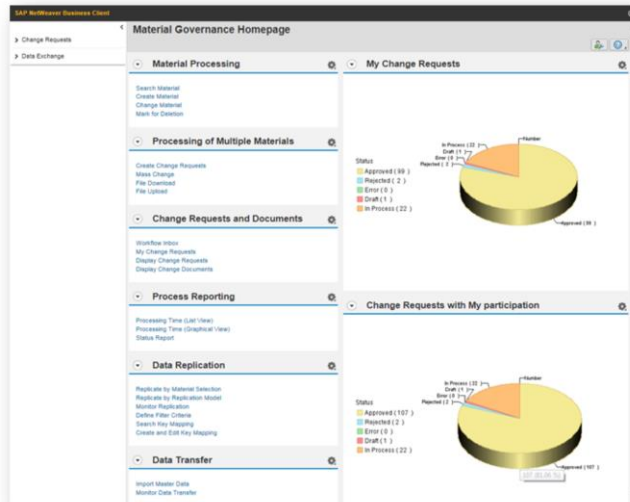
- Users maintained master data in different systems
- Different transactions were necessary to execute the maintenance tasks (change, create, delete)
- Planned changes were not transparent to other systems

- **Business value**

- The new work center for material master data governance bundles together all activities for material master data maintenance.
- Central deployment and authorization ensures governance
- Local adaptation and personalization possible

User-Centric Principle

Homepage with PFCG Role SAP_MDGM_MENU_06



© 2016 SAP SE or an SAP affiliate company. All rights reserved.

Early Knowledge Transfer

Customer

19

When starting the Homepage for Material Governance via the Netweaver Business Client, you see the entry screen above.

The business processes are grouped in main areas

- Material processing
- Processing of Multiple Materials
- Change Requests and Documents
- Process Reporting
- Data Transfer
- My Change Requests (Graphic)
- Change Requests with My participation (Graphic)

If you start with Change Requests, a personal worklist is displayed including all change requests where you are involved. You can adjust this list via filters, groupings, etc. to your needs.

Enterprise Portal Integration:

No new delivered portal content with MDG 8.0 for MDG-M

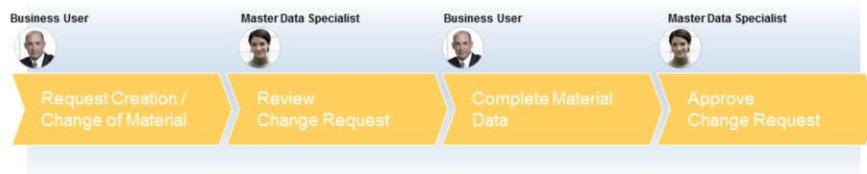
Instead, the new PFCG Roles (e.g. SAP_MDGM_MENU_06) can be transferred to the portal

Governance Principle

Change Requests With Built-In Approval Process

This function allows you to implement an integrated process of master data governance that starts with a request to create or change a material, continues with processing and commenting by different expert users and stakeholders, and finishes with approval or rejection by authorized experts.

All processing steps are supported by workflow.



Customer business challenges

- Requests and reasons for new or changed master data had to be communicated by phone or e-mail to master data specialists. They had to gather any further information in a time-consuming process from other experts and stakeholders. Follow-up processing by certain specialists had to be triggered manually. An approval process with a two man rule could not be assured.

Business value

- The approval process for maintenance of master data enables:
 - Fulfillment of compliance requirements
 - Better data quality
 - Faster and transparent decisions
 - Relevant information in one place

Workflow Principle

Standardized but Flexible Processes Incl. Rules Management

The data maintenance activities are bundled via a change request that is linked to a workflow. The workflow can be linear or distributed. Besides automated steps, ad hoc steps, delegation etc. can be used. The standard SAP Business Workflow is enhanced with a rule-based engine; therefore changes in responsibilities and in processes can be reflected immediately and with ease.



Customer business challenges

- Processes were unstructured and neither guided nor controlled
- High effort in data collection, change processing and issue resolution
- High TCO and low flexibility if hard-coded semi-automated processes are used

Business value

- Build-in guidance, monitoring and reporting using a standard workflow tool provides process and data governance
- Integrated tool for keeping data, process and issue resolution in one place
- Enhanced flexibility by integrated rules management

Architecture Principle

Staging Area to Maintain Inactive Versions of Material Data

As long as the change request for the creation or change of a material is in progress, the attributes of the material are stored in an inactive version that exists (in the case of a change) in parallel to the active version. When the change request is approved then the data is written as the active version to the master data tables and this new or updated record is then available for business processes.



Customer business challenges

- Several users from different departments are involved in the creation of a material item
- Material items were immediately available in the system as soon as the first user saved the entries
- Many incomplete material items and partial changes could exist
- Planned changes were usually not visible to the business

Business value

- Incomplete material (changes) are stored in an inactive version so the material can only be used in business processes when it is approved (better process quality)
- Rejected change requests don't create a material number in SAP ERP (better data quality)
- Planned changes are visible to the concerned user groups (better transparency)

Architecture Principle

Handling of Active and Inactive Data

To enable the master data governance process MDG separates

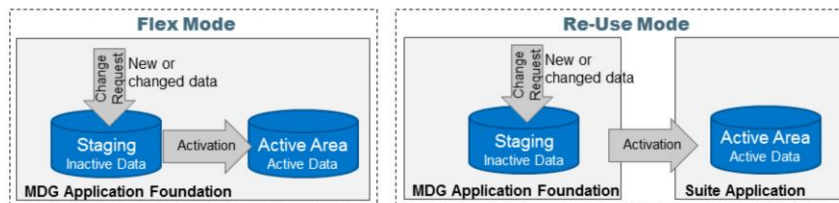
- Active data – Ready to be used by applications, can be distributed to client systems
- Inactive data – Not yet approved, currently part of a change request

Accordingly there are two separate storage locations

- Active area – Holds the active data
- Staging – Holds the data during the change request processing before approval

For optimal integration into SAP Business Suite MDG allows two modes

- Re-Use active area (re-use mode) : Existing structures of applications are used. For example, MDG for material makes use of the MARA table in ECC
- Generated active area (flex mode) : Tables as defined in the MDG data model are used to store active data.



© 2016 SAP SE or an SAP affiliate company. All rights reserved.

Early Knowledge Transfer

Customer

23

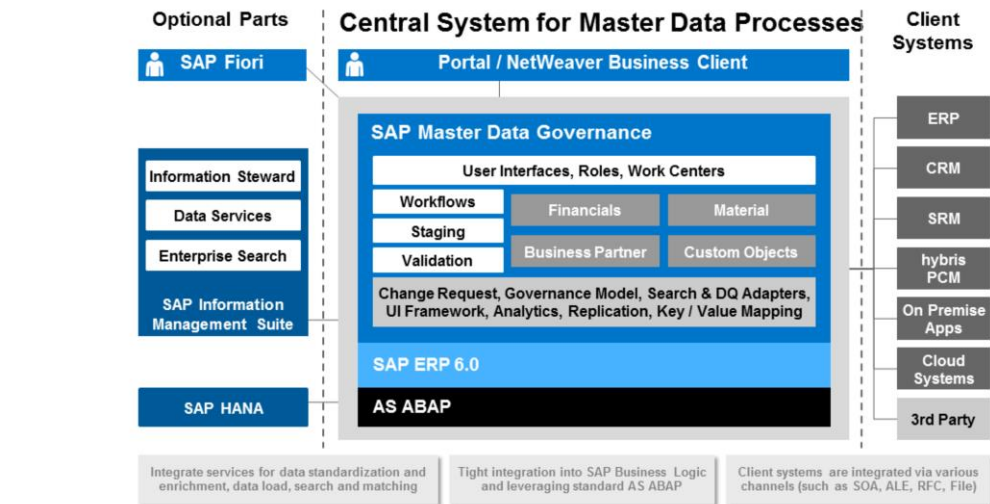
The active version of one entity (e.g. one material master record) holds the approved active data and is stored in the active area. Based on this version the distribution takes place and in case of reuse and integrated MDG also the operations are executed on the active data.

MDG for material data uses the re-use mode for integration into the business suite. This has some advantages

- Reuse of existing business logic for the backend validation
- Easy integration of existing customer enhancements
- Reuse of backend authorization
- Existing distribution mechanisms can be used (ALE/IDoc, CIF, Middleware)

SAP Master Data Governance

High-level Architecture Overview



© 2016 SAP SE or an SAP affiliate company. All rights reserved.

Early Knowledge Transfer

Customer

24

This picture shows an overview of the general architecture of MDG. We will not go into detail here, I just wanted to point out a few things here:

Same as the main SAP Business Suite components, MDG is built on top of the Application Server for ABAP (AS ABAP). Parts of MDG are in a software layer “underneath SAP ERP” and can be re-used by ERP’s applications. Other parts of MDG are in a software layer “on top of ERP” and can re-use existing ERP coding, like for example the build-in validations for ERP-related master data do. MDG makes use of components of SAP Information Management portfolio like for example for fuzzy searches it can use Enterprise Search or it leverages SAP Data Services for address validation.

MDG provides HTML-based user interfaces based on WebDynpro for ABAP as well as roles that can be accessed either through the SAP NetWeaver Portal or through the NetWeaver Business Client.

MDG’s Data Replication Framework manages the distribution of master data to SAP and non-SAP systems using technologies like ALE IDoc or Enterprise Services.

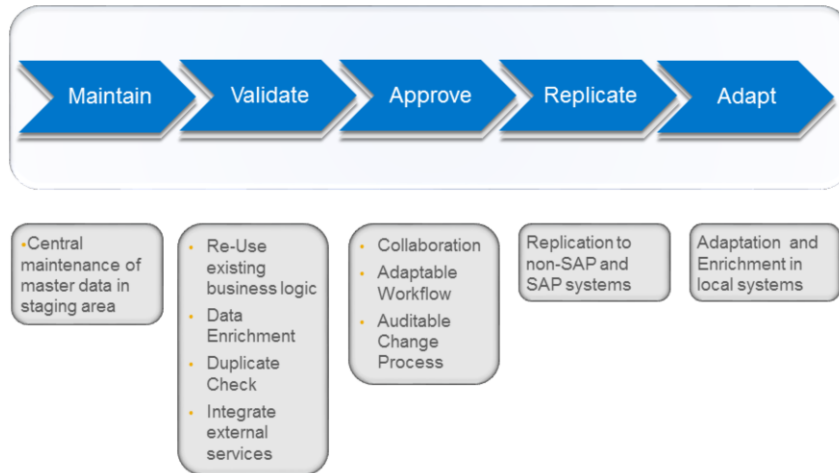


Agenda

1. Current Reality
2. Need for the Future
3. SAP Master Data Governance Capabilities
4. Architecture
- 5. Current Solution**
6. Benefit from the Solution



SAP Master Data Governance – Process Flow



Material Processes

Search Functionality

- In their work centers, users can search for materials that are stored in the active or in the staging area.
- Search criteria combine material attributes and classification
- Search uses Enterprise Search (TREX) and/or SAP HANA-Based Search or other alternative search provider.
- The user can start a change request for single or multi records processing directly from the results list.



Customer business challenges

- Search results only took operational or active data into account, not the inactive data in the staging area related to planned or future changes
- Business activities could not be directly triggered from a search result list

Business value

- Comprehensive search functionality simplifies daily work
- Integration into change request processing enhances productivity
- Including inactive data from staging area reduces parallel change processes

Search – First Step to Pristine Data Quality

Search is the initial action, if a user wants to “do something” with the data.

Several Flavors Supported via Search Provider Concept

Standard provider: Enterprise Search incl. Fuzzy Search

Customers can integrate their own search providers

Material Processes

Create Material With Template, Initial Screen

The screenshot shows the 'Material: New' initial screen in SAP. It is divided into two main sections: 'Material' and 'Material (Template)'. The 'Material' section has fields for Material (P-3000), Plant (0001), Storage Location (0001), Sales Organization (0001), Distribution Channel (01), Warehouse Number (001), and Storage Type (001). The 'Material (Template)' section has fields for Material (P-100), Plant (0003), Storage Location (0002), Sales Organization, Distribution Channel, Warehouse Number, and Storage Type. Below these sections are 'View' and 'Version' tabs. The 'View' tab shows 'Change Request Type' as MAT01. The 'Version' tab shows 'Change Request' as 0. A 'Continue' button is at the top left.

- Copy from a Template: Enter Material, and if necessary a Plant, Storage Location, Sales Organization and Distribution Channel, Warehouse Number and Storage Type
- If you specify a Change Request as template than inactive data of the material is copied
- Only specified organizational levels of template are copied. Otherwise no organizational level is copied.

You can use *Create Material* to create a change request for material creation. This change request goes through the predetermined workflow process and when complete results in the creation of a new material in the active area.

Process Steps:

- In the Material field enter a material ID or leave it blank and the system will create a temporary ID that will be changed to the standard ID when the change request is approved.
- Enter change request details like description, and further details such as priority, due date, and reason. You can also add comments and attach or link relevant files.
- Enter details for new material
- Submit your change request to trigger the proceeding workflow steps.

The new change request for material creation is created and sent for processing according to the rule-based workflow you have assigned to the create material process. Usually this means that it passes to a Master Data Steward who reviews the changes. During this time the new material is stored in the staging area.

Material Processes

Create Material Overview Page

- Several actions are possible for processing a change request (Customizing)
- Workflow is controlled by the BRFplus decision tables when using the Rules based Workflow template or controlled by the Classic Workflow template
- Dedicated users or organizational units are derived for further processing
- Delivered template configuration: BS_MAT_OVP_06

You can use *Create Material* to create a change request for material creation. This change request goes through the predetermined workflow process and when complete results in the creation of a new material in the active area.

Process Steps:

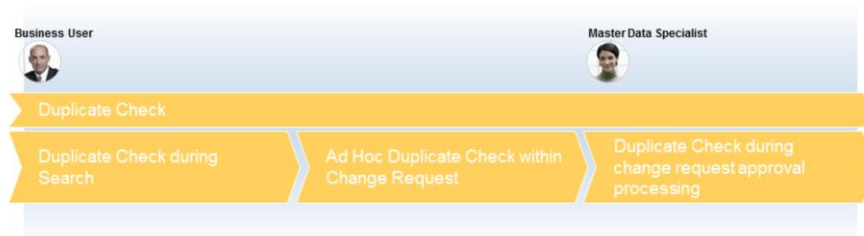
- In the Material field enter a material ID or leave it blank and the system will create a temporary ID that will be changed to the standard ID when the change request is approved.
- Enter change request details like description, and further details such as priority, due date, and reason. You can also add comments and attach or link relevant files.
- Enter details for new material
- Submit your change request to trigger the proceeding workflow steps.

The new change request for material creation is created and sent for processing according to the rule-based workflow you have assigned to the create material process. Usually this means that it passes to a Master Data Steward who reviews the changes. During this time the new material is stored in the staging area.

Material Processes

Duplicate Check Using Enterprise Search

A basic duplicate check is offered to avoid the creation of already existing materials. The information on potential duplicates is provided during when checking or submitting the change requests. It can be switched on or off per workflow steps.



Customer business challenges

- Users often created a new material item even if the material master already existed in the system. This resulted not only in bad data quality, but also generated unnecessary cost since every master data record has its price tagged during its lifecycle.

Business value

- The duplicate check improves the data quality as well as the decision and business process quality in the connected systems
- Avoiding duplicates as early as possible saves time and money in the master data process

Material Processes

Display Material

Material: P-021014-03; Plant: 0001

Save Cancel Edit Expand all Collapse all Assignment Block

Change Management

Basic Data Edit Details

General Data	Grouping
Material: P-021014-03	External Material Group: 0000... Screwdrivers
Base Unit of Measure: EA each	Product Hierarchy: 0000... NORM Standard Item
Material Type: FERT Finished Product	Generic Item Group Category: NORM Standard Item
Industry Sector: M Mechanical engineering	Material Group Packaging: ...
Material Group: 02 Material group 2	Design Data
Old Material Number: P-020114	Lab/Office: 001 Laboratory 1
Authorization Group: EM	Industry Standard Descript...
Cross-Plant Material Status: 01 Blocked for procurement since 30.12.2014	Basic Material: Steel
Batch Management: ☐ No	Production/Inspection Mem...
Serialization Level: ☐ No	CAD Indicator: ☐ No
Approved Batch Record Re... ☐ No	Configuration
Division: 01 Product Division 01	Cross-Plant Configurable M...
Dimension	Material Configurable: ☐ No
Net Weight: 0.000 KG Kilogram	Environment
Gross Weight: 0.000 KG	Environmentally Relevant: ☐ No
Length: 10.000 MM Millimeter	Dangerous Goods Indicator...
Width: 0.000 MM	GTIN/EAN
Height: 0.000 MM	EANUPC:
Volume: 0.000	EAN Category:
ATP	EAN Variant:
Size/Dimensions:	
Product Allocation:	

- Display material via WebDynpro Application Change Material
- Or display material via WebDynpro Application Search Material and click on the Material Number link
- Switch with the edit mode to create a new change request or to existing change request

Material Processes

Change Material With Template, Initial Screen

- Copy from a Template: Enter Material, and if necessary a Plant, Storage Location, Sales Organization and Distribution Channel, Warehouse Number and Storage Type
- If you specify a Change Request as template than inactive data of the material is copied
- Only specified organizational levels of template are copied. Otherwise no organizational level is copied.

You can use *Change Material* to create a change request for changing a single material. This change request goes through the predetermined workflow process and when complete results in the change of the selected material in the active area. Creating the change request copies the material to the inactive area. A material can only be assigned to one active change request at a time, so no new change requests can be created for a material that is already in the inactive area.

Process Steps:

- Select a material: Enter a material number in the Material field or use the input help to choose one. Click the Start button to continue.
- Enter change request details: In the Change Request section, enter a description for your change request in the Description field. You can enter details for your change request such as priority, due date, and reason. You can also add comments and attach relevant files if you wish.
- Enter changes: Enter the changes you want to make to the material and click the Submit button to submit your change request.

Your new change request is created and sent for processing according to the rule-based workflow you have assigned to the change material process. Normally this means that it passes to a Master Data Steward who reviews the changes.

Once the change request has been approved and then technical validation has taken place the changed material data replaces the existing data in the active area. The material is no longer available in the inactive area.

Material Processes

Change Material

- Review Changes by showing the Change Documents
- Several actions are possible for processing a CR (Customizing)
- Workflow is controlled by the BRFplus decision tables when using the Rules based Workflow template or controlled by the Classic Workflow template
- Dedicated users or organizational units are derived for further processing
- Delivered hierarchical template configuration: BS_MAT_OVP_06H

You can use *Change Material* to create a change request for changing a single material. This change request goes through the predetermined workflow process and when complete results in the change of the selected material in the active area. Creating the change request copies the material to the inactive area. A material can only be assigned to one active change request at a time, so no new change requests can be created for a material that is already in the inactive area.

Process Steps:

- Select a material: Enter a material number in the Material field or use the input help to choose one. Click the Start button to continue.
- Enter change request details: In the Change Request section, enter a description for your change request in the Description field. You can enter details for your change request such as priority, due date, and reason. You can also add comments and attach relevant files if you wish.
- Enter changes: Enter the changes you want to make to the material and click the Submit button to submit your change request.

Your new change request is created and sent for processing according to the rule-based workflow you have assigned to the change material process. Normally this means that it passes to a Master Data Steward who reviews the changes. Once the Master Data Steward has approved the change request and technical validation has taken place the changed material is moved into the active area.

Once the change request has been approved and then technical validation has taken place the changed material data replaces the existing data in the active area. The material is no longer available in the inactive area.

Material Processes

Change Material Type

Material: P-021014-03; Plant: 0002; Storage: 0001

Save Submit Cancel Edit Expand all Collapse all Switch to Active Version Check Assign

Change Request

Change Management

Basic Data Edit Details

General Data

Material: P-021014-03

* Base Unit of Measure: EA each

* Material Type: FERT Finished Product

Industry Sector: M Mechanical engineering

Material Group: 02 Material group 2

Old Material Number: P-020114

Authorization Group: EM

Cross-Plant Material Status: 01 Blocked for procurement (30.10.2014)

Batch Management: No

Serialization Level: No Serialization within the stock material number

Approved Batch Record Required Indica...: No

Division: 01 Product Division 01

Dimension

Net Weight: 0.000 KG Kilogram

Gross Weight: 0.000 KG

Grouping

External Material

Product Ho

Generic Item Group Co

Material Group Packaging Ma

Design Data

Lat

Industry Standard Des

Production/Inspection

Page Format of Production

CAD In

Configuration

Cross-Plant Configurable M

Material Conf

Environment

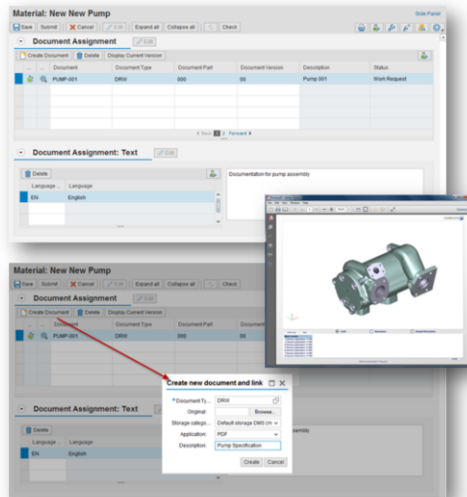
- With the Material Change UI it is possible to change the Material Type
- Prerequisite: Authorization M_MATE_MTA

You can use *Change Material* to create a change request for changing the material type a single material if you have the authorization to do so.

Material Processes

Document Management System (DMS) Integration

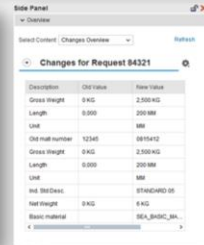
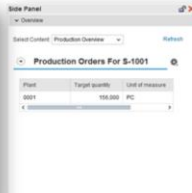
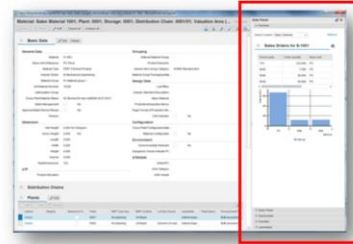
- Display, edit and delete links to document info records
- Maintain long text information for material link
- Easy creation of new documents (and material link)
- Display document info record dependent on user/system environment
 - Using PLM Web UI
 - Using SAP GUI HTML
 - Using SAP GUI (NWBC for desktop only)



Material Processes

BCV Side Panel Content for MDG Material

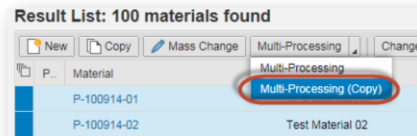
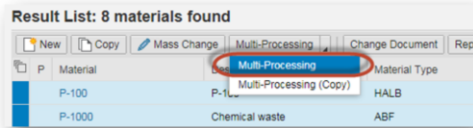
- Sales Overview: Sales orders created for the current material master.
- Production Overview: Production orders created for the current material master.
- Purchasing Overview: Purchase orders created for the current material master.
- Changes Overview: Changes raised by the current MDG change request.



Material Processes

Multiple Record Processing

- Change selected materials when navigating from the search UI to the multiple record processing application
- Copy multiple materials with sub-objects when navigating from the search UI to the multiple record processing application
- Combined maintenance of different entity types
- List and Form UIBB's are supported
- Example UI configurations and customizing delivered by business configuration sets



Material Processes

Multiple Record Processing

- Add, copy and change materials, language-dependent texts and organizational data in a change request
- Delete, remove and discard changes possible, with highlighting
- Copy with reference
- Create MuRP Change Request without selecting objects
- Add existing objects with New Button
- Several operators available in context menu (e.g. select rows with identical value)
- Highlighting changes with different colors
 - Saved changes
 - Current changes

Materials

Find and Replace | Add | Copy | Discard Changes | Remove

Changes	Material	Material T...	Material S...	Base Unit...	Base Unit...
	37052	PERT	M	EA	EA
	37053	PERT	M	EA	EA

Change Request

Find and Replace | Highlight Changes

Material	Material Type	Industry sector	Basic Unit of Me...	Material Group	Old material nu...	Current mat. val.	Valid from
P-070813-01	Holz	M	EA	BO2		40	
P-070813-02	Holz	M	EA	BO2	P-070813-02	40	
P-070813-03	Holz	M	EA	BO1	P-070813-03	40	
P-070813-04	Holz	M	EA	BO2		40	

Planning

Find and Replace | Highlight Changes

Material	Material Type	MSP Code	MSP profile	MSP group	Schedule	Lot size	Fixed size	Minimum	Maximum
P-070813-01	BO2	MD	001	0001	000	30	0.000	0.000	0.000
P-070813-02	BO2	MD	001	0001	000	EX	0.000	0.000	0.000
P-070813-03	BO2	MD	001	0001	000	EX	0.000	0.000	0.000
P-070813-04	BO2	MD	001	0001	000	EX	0.000	0.000	0.000

Material Group

Old material nu...

BO2	Select Rows with Identical Value
BO2	Find and Replace
CTG	Copy Value to Selected Rows
BO2	Display Quick Help
	More Field Help...
	Technical Help...

Material Processes

Mass Change

- Optionally, create a change request and submit it (typically w/o materials assigned, yet)

- Second, start the mass change and select your relevant materials by Define Selection (Step 1) and Refine Selection (Step 2)

- Step 3: Replace one or more fields with one dedicated value

- Step 4: You can review the results and then execute the change. If not yet done, you now need to assign an open mass change request that is assigned to you, or you can create a new change request

Mass Change Material: Step 1 (Define Selection)

Progress: 1 Define Selection | 2 Refine Selection | 3 Define Values | 4 Check and Execute

Selections:

Material: To:

Authorization Group: To:

Old material number: To:

Mass Change Material: Step 3 (Define Values)

Progress: 1 Define Selection | 2 Refine Selection | 3 Define Values | 4 Check and Execute

Changeable Attributes:

Attribute	Replace With	Inherit
Owner	01	☐
Storability Factor		☐
Transportation Group	0001	☐

Select or Create Change Request

Change Request:

The mass change allows to replace for the selected materials a new value per field. Updating multiple materials is a two step process; you first create a change request for multiple materials, then you use the *Mass Change* to enter the details for the changes you want to make.

- Open the *Mass Change* screen: Select the *Mass Change* link from the *Material Governance* home page.
- Select materials to include: Use the *Define Selection* to choose the materials you want to include in your change request. Click the *Next* button to continue.
- Refine your selection: Use the *Refine Selection* to review your chosen materials and remove any materials you do not want to include in your change request. Note that if a material you have selected already exists in the staging area, this change request overwrites it. Click the *Next* button to continue.
- Define values: Choose the attributes you wish to change and the value you wish to set them to. To do this, enter the new value in the *Replace With* column of the attribute you want to change. This value is set for the attribute in each material you selected. Note that only attributes available for mass change are visible on this screen. Click the *Next* button to continue.
- Check and execute: Review your changes. If you are happy with them, click the *Execute Changes* button to submit your change request.
- If not yet done, you now need to assign an open mass change request that is assigned to you

Material Processes

Mark Material for Deletion

Delete Material

[Continue](#)

Material

* Material: P-100

Organizational Levels

Plant: 0001

Storage Location: 0001

Sales Organization: 0001

Distribution Channel: 01

Warehouse Number:

Storage Type:

View

Change Request: 0

* Change Request Type: MAT06

- Mark a single material for deletion
- Or set the flag on a org level
- A change request is triggered and data is copied into the staging area

Delete Material: P-100; Plant: 0001; Storage: 0001; Distribution Chain: 0001/01

[Save](#) [Submit](#) [Cancel](#) [Edit](#) [Expand all](#) [Collapse all](#) [Switch to Active Version](#) [Check](#)

Change Request [Edit](#)

General **Notes** **Attachments**

General Data

Change Request ID: 82173

* Description: Deletion flag for Storage Location 0001

Priority:

Due Date:

Process Data

Status: Changes to Be Executed

Current Work Item: New Change Request

Created OnBy: 06.09.2013 10:48:52 [View](#)

Changed OnBy: 06.09.2013 10:49:03 [View](#)

Basic Data

Distribution Chains

Plants

Plant: Storage Locations [Edit](#)

[Discard](#)

Staging	Marked for Deletion	Storage Location	Storage Location MRP Indicator Description
☒	☐	0001	Storage location stock also planned at plant level

- Workflow is triggered and processed the same way as for the other processes
- The deletion is limited to the material entity. Currently no remote checks about reuse of the material is done.

The service to mark a material for deletion allows to plan this via MDG.

Process Steps:

- Select a material: Enter a material ID in the Material field or use the input help to search for a material ID. To continue, click the Start button.
- Enter change request data: In the *General Data* tab enter the change request description, priority, due date, and reason.
- Set the delete option and submit: Set the value for the radio button *DF at Client Level* to "Yes". To create the change request, click the *Submit* button.

The system creates a change request for deleting the chosen material. The change request enters the appropriate workflow and once approved results in the material being deleted from the active area (if not limited by other business rules).

Support of Parallel Change Request

Multiple Change Requests per Material at the Same Time

Creation of material with initial Change Request

One Change Request for create material

Parallel Change Requests for subsequent individual processing

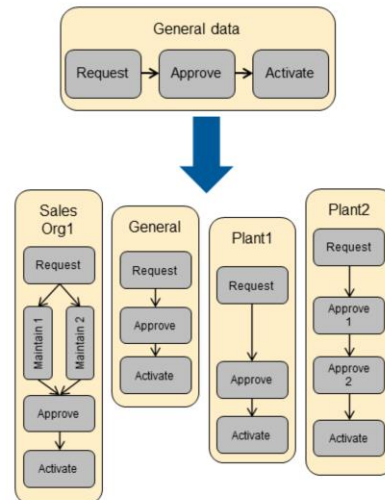
- Parallel Change Requests for extending material
- Extend material to different organizations or change existing data in parallel to extend the material

Activation for general data and per organizational unit directly after approval

Workflow creation for each selected organizational unit, activation after approval

Lock on entity level (interlocking)

- Instead of lock on object level only
- Change request type determines the maintainable fields



Interlocking

MDG Framework Capabilities

Common Services of MDG for Material

Solution Strategy

- Integrated Object Model
- Deliver Out-of-the-Box
- Re-Usability & Extensibility of Data Model & User Interface
- Data Replication
- Governance, Collaboration & Data Quality

Change Requests for Material with built-in approval process (see previous slides)

- Central Work Center for the Maintenance of Materials
- Search, Create, Display, Change, Mark for Deletion and Mass Change Material
- Staging Area: Material Data is only created when Change Request is approved

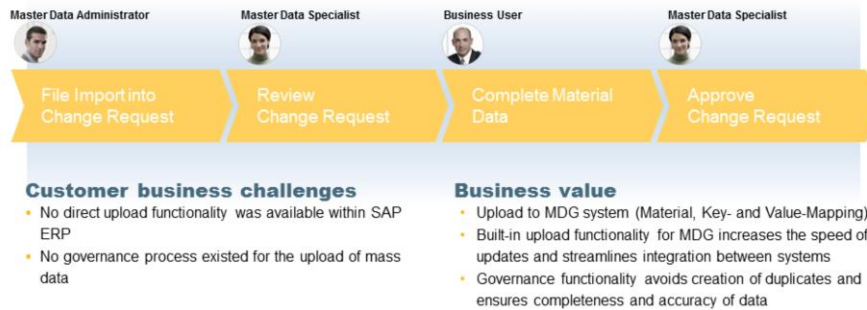
Common services of change request processing (explained on the next slides)

- Import Master Data
- Key- and Value-Mapping
- Worklists (My Change Request, Display Change Requests, Change Documents)

Import Master Data

Common Services of MDG for Material

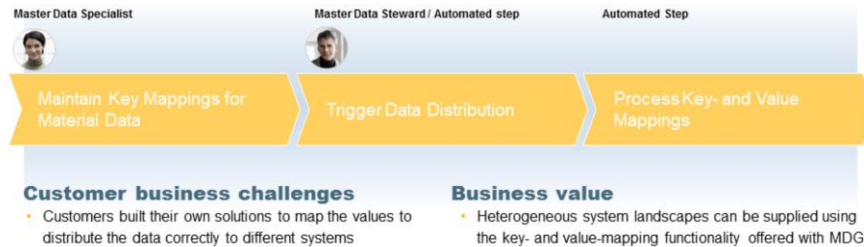
Material master data, Key- and Value-Mapping can be uploaded to the MDG system from an XML file using the MDG Import Master Data transaction. The imported material master data can be adjusted or modified in a change request. Only when the change request is approved is data written to the SAP ERP material master data (which is the active area). If data quality is good, the data can also be written directly into the active area.



Key- and Value Mapping

Common Services of MDG for Material

This feature supports the replication of data into systems with non-harmonized customizing or with heterogeneous keys for material. For example, in cases where the same material has a different material number on the master data governance (MDG) hub than on a client (key mapping), or similar material groups have a different code in hub and clients (value mapping).



Worklists

Common Services of MDG for Material

- My Change Requests

My Change Requests

Close | Print Preview | Side Panel

Name: Elke

Change Requests

Show: Created by Me | From: 30.07.2013 | To: 06.08.2013 | Go | View: [Standard View]

Print Version | Export | Workflow Log

Change Request	Description	Status	Changed On	Changed By	Draft
633	P-070813-12 635	Final Check Approved	15.08.2013	Andrea Becker	☐
577	P-070813-03	Final Check Approved	12.08.2013	Rita Mayer	☐
574	Change	Final Check Approved	12.08.2013	Andrea Becker	☐

- Change Documents

Display Change Documents

Close | Entity Type: Material

Display: Attribute Changes | View: EM | Print Version | Export | Descriptions

Value	User Name	Date	Time	Change	Attribute	Old Value	New Value
R00 601 909 314	Andrea Buch	08.07.2013	13:14:17	New	Material Group		DUMMY
		09.07.2013	12:53:52		X-dist chain status		01
					Valid from	00.00.0000	10.07.2011
	Andrea Becker	10.07.2013	11:15:21	Deleted	Material Group		DUMMY
		10.01.50		New	Old material number		601909314
					Material Group		DUMMY

- Display Change Requests

Display Change Requests

Continue | Close

Change Request: [] To: []

Type of Change Request: [] To: []

Status of Change Request: [] To: []

Edition: [] To: []

Due Date: [] To: []

Priority: [] To: []

Reason: [] To: []

Reason for Rejection: [] To: []

Created By: [] To: []

Created On: [04.09.2013] To: [06.10.2013]

Changed By: [] To: []

Changed On: [] To: []

Finalized By: [] To: []

Finalized On: [] To: []

No Finalized Change Requests: ☒

SAP Master Data Governance for Material Data

Integrated, Centralized Product/Material Data for Process Excellence



Material Processing

- Workflow-driven creation and change process for Materials
- Change requests steer the maintenance & approval process of new or changed data
- Document changes for audits
- Work lists organize due changes
- Adaptable user interfaces and validations can be different per workflow step if needed

Distribution

- Changes are kept inactive until approval
- They can then be distributed to operational group or local systems via DRF using ALE

Scope of Master Data

- Search, Display, Change, Creation of single Material Master Data
- Mass and multi processing of changes
- File upload for initial load, system consolidation
- Broad out-of-the-box coverage of material attributes – incl. central and classification data as well as organizational unit data

Material master data is a very complex thing. In some organizations, more than 20 people need to contribute to the maintenance of a single material master. Especially for introduction of new products many companies are struggling to ensure that all that information is maintained properly and completely before the new material is used in the various business processes. To make it even more complex, different data has to be complete in order to use the material in production planning, or in sales order processing, or in shipments. This complexity demands for a solution that organizes the process of the data maintenance across multiple users and allows to absolutely avoid that data is being used before it is ready.

As discussed in more detail on the last slide, MDG will help these organizations to manage exactly this complexity. Customers can use the standard MDG content as a very good starting point and then create their specific processes, typically defining multiple Types of Change Request in MDG for various usage scenarios.



Agenda

1. Current Reality
2. Need for the Future
3. SAP Master Data Governance Capabilities
4. Architecture
5. Current Solution
- 6. Benefit from the Solution**



SAP's Understanding of a State-of-the-Art

Master Data Governance Solution

**Ensure Master
Data
Governance**



Solution Strategy

- Deliver process-centric solution
- Workflow closely integrated to a framework for defining business rules
- Full Integration to SAP Systems and integration services for non-SAP
- Deliver roles on a web based UI
- Deliver a Data Model proven by several applications & industries
- Reuse of existing processing logic and out-of-the box data replication mechanisms

Basic Principles

- User-centricity: Central work center for the maintenance of materials
- Governance: Change requests with built-in approval process
- Workflow: standardized but flexible processes incl. rules management and tracking
- Architecture: Staging area to maintain inactive versions of material data

Common Services

- Search & Display
- Create, Change, Multi-Record Processing, Mass Change, Parallel Change Requests, Mark for Deletion
- Work lists (My Change Request, Display Change Requests, Change Documents)
- Duplicate Check
- Data Replication with Key- and Value-Mapping
- File Upload / Initial Load
- Extensibility of Data Model & User Interface
- Deployment on top of operational ERP or as standalone hub

We are approaching the end of this session. This slide is just trying to summarize why we think that we have a very unique offering with SAP MDG, that is unrivalled in the market when it comes to central MDM for SAP-centric data.

In short:

- MDG is an out-of-the-box solution integrated into the customers SAP Business Suite context.
- It provides all the needed ingredients for a collaborative governance solution that ensures high data quality through being both: process-centric as well as user-centric.
- It comes with a framework and all the common services that companies will need to optimize the solution to their needs.
- And all this comes at a low TCO for the customer, as MDG behaves like the SAP ERP application they are already using today.

Benefits From SAP Master Data Governance



- Up-front, central creation of master data enables high data quality and leads to significant business improvements.
- Out-of-the-box, process-centric applications provide centralized governance for master data domains such as financial data, supplier data, customer data, material data.
- Native integration with SAP Business Suite using existing business logic and infrastructure leads to low TCO.
- SAP Master Data Governance provides flexibility, from configurable workflows and extensible data models, up to custom built applications.

This slide is trying to summarize the value proposition of SAP MDG and trying to call for action:

- Wherever possible, organizations should strive for central master data management for their most critical data. Through this they can eliminate the root causes for business problems resulting from bad data.
- MDG is the out-of-the-box solution to their needs.
- MDG is built to be integrated into the customers SAP landscape.
- Although being out-of-the-box, MDG is the platform that provides the flexibility their organization needs.

Lesson Summary



You should now be able to

- Explain the goal of Master Data Governance for Material
- Describe the strategy and the features of the current solution
- Outline the architecture of Master Data Governance for Material
- Describe the governed maintenance process for material master data with Change Request and workflow

No part of this publication may be reproduced or transmitted in any form or for any purpose without the express permission of SAP SE or an SAP affiliate company.

SAP and other SAP products and services mentioned herein as well as their respective logos are trademarks or registered trademarks of SAP SE (or an SAP affiliate company) in Germany and other countries. Please see <http://global12.sap.com/corporate-en/legal/copyright/index.epx> for additional trademark information and notices.

Some software products marketed by SAP SE and its distributors contain proprietary software components of other software vendors.

National product specifications may vary.

These materials are provided by SAP SE or an SAP affiliate company for informational purposes only, without representation or warranty of any kind, and SAP SE or its affiliated companies shall not be liable for errors or omissions with respect to the materials. The only warranties for SAP SE or SAP affiliate company products and services are those that are set forth in the express warranty statements accompanying such products and services, if any. Nothing herein should be construed as constituting an additional warranty.

In particular, SAP SE or its affiliated companies have no obligation to pursue any course of business outlined in this document or any related presentation, or to develop or release any functionality mentioned therein. This document, or any related presentation, and SAP SE's or its affiliated companies' strategy and possible future developments, products, and/or platform directions and functionality are all subject to change and may be changed by SAP SE or its affiliated companies at any time for any reason without notice. The information in this document is not a commitment, promise, or legal obligation to deliver any material, code, or functionality. All forward-looking statements are subject to various risks and uncertainties that could cause actual results to differ materially from expectations. Readers are cautioned not to place undue reliance on these forward-looking statements, which speak only as of their dates, and they should not be relied upon in making purchasing decisions.